

Physical and Mathematical Requirements for an Accurate 3D Billiards Simulation

Executive summary

Accurate 3D billiards/pool simulation is a coupled **rigid-body dynamics + intermittent contact/impact** problem with strong dependence on (i) frictional transitions (sliding→rolling), (ii) short-duration impulses (ball-ball and ball-cushion), and (iii) geometry-driven capture/rejection in pockets. Regulation equipment specifications constrain key geometric parameters—ball size/mass, cushion nose height, pocket mouth/shelf/cut angles—while experimental work provides realistic ranges for friction and restitution that dominate macroscopic behavior (stop distance/time, rebound angles, spin survival). ¹

A robust simulation needs **two layers of physics**:

1. **Between impacts** (continuous dynamics): Newton–Euler motion with a **state-dependent contact model** at the cloth interface (sliding friction, rolling resistance, and spin decay). The key transitions can be modeled analytically (Coulomb sliding) and matched to measured friction ranges for billiards cloth. ²
2. **At impacts** (discontinuous dynamics): an impulse-based collision response (with Coulomb friction and rotational coupling) is typically sufficient for real-time; a compliant (Hertz/Mindlin-type) contact model can be used when you want contact duration, peak forces, and speed-dependent restitution. Canonical viscoelastic and frictional impact literature supplies implementable force laws and restitution dependence on speed. ³

The largest practical sources of visible error in game-like simulations are:

- **pocket geometry not matching spec** (changes acceptance/rejection dramatically), ⁴
- **wrong rolling resistance** (stop distances and “table speed” feel wrong), ⁵
- **missing tangential impulses** during collisions (spin/throw/deflection wrong), ⁶
- **numerical tunneling or energy gain** from unstable time stepping, ⁷
- **spin decay modeling** (especially sidespin) inconsistent with measured angular deceleration. ⁸

Modeling scope and assumptions

This report treats “accurate simulation” as matching: (a) trajectories and stopping distances on cloth, (b) post-collision linear and angular velocities for ball-ball and ball-cushion impacts, and (c) pocket acceptance regions as a function of approach speed/spin/angle. Table model is unspecified, so defaults assume a regulation-style rectangular pool table compliant with World Pool-Billiard Association ⁹ recommended equipment specifications, with alternative values from Billiard Congress of America ¹⁰ where relevant. ¹¹

Variants are included as parameterizations:

- Snooker: ball diameter 52.5 mm (tight tolerance), and pocket openings specified by templates rather than explicit dimensions in the published rulebook. ¹²
- Carom: no pockets; specified playing surface and cushion height are defined in Union Mondiale de Billard ¹³ rules. ¹⁴

Coordinate conventions used throughout:

- World frame: table bed is plane $z = 0$, gravity $\mathbf{g} = (0, 0, -g)$.
- Each ball i : center position $\mathbf{x}_i$, linear velocity $\mathbf{v}_i$, angular velocity $\boldsymbol{\omega}_i$, mass m_i , inertia tensor $\mathbf{I}_i = I_i \mathbf{1}$ (spherically symmetric). ¹⁵

Continuous dynamics on cloth: linear motion, angular motion, friction, and spin decay

Newton–Euler equations ready to implement

For each ball i (full 3D):

$$\begin{aligned}\dot{\mathbf{x}}_i &= \mathbf{v}_i, & m_i \dot{\mathbf{v}}_i &= \mathbf{F}_{\text{ext}} + \mathbf{F}_{\text{cloth}} + \mathbf{F}_{\text{air}} \\ \mathbf{I}_i \dot{\boldsymbol{\omega}}_i &= \boldsymbol{\tau}_{\text{ext}} + \boldsymbol{\tau}_{\text{cloth}}\end{aligned}$$

To update orientation, use quaternion q_i with:

$$\dot{q}_i = \frac{1}{2} q_i \otimes (0, \boldsymbol{\omega}_i)$$

This is the standard rigid-body formulation used in impulse/constraint methods; see impulse-based and time-stepping treatments for contact/friction. ⁷

Cloth contact kinematics: slip velocity at the contact point

Let $\mathbf{n} = (0, 0, 1)$ be the table normal and the contact point in body coordinates be $\mathbf{r} = -R\mathbf{n}$. The instantaneous velocity of the contact point is:

$$\mathbf{v}_c = \mathbf{v} + \boldsymbol{\omega} \times \mathbf{r}$$

The **slip** (relative tangential velocity) is the projection to the plane:

$$\mathbf{u} = \mathbf{v}_c - (\mathbf{v}_c \cdot \mathbf{n})\mathbf{n}$$

Pure rolling (no slip) means $\mathbf{u} = \mathbf{0}$. This is the key state variable for transitioning between sliding and rolling. ¹⁶

Sliding friction model (dominant immediately after cueing and after many impacts)

A practical Coulomb sliding law:

$$\mathbf{F}_{\text{slide}} = -\mu_s N \frac{\mathbf{u}}{\|\mathbf{u}\| + \varepsilon_u}$$

where $N \approx mg$ when the ball remains on the bed, and ε_u is a small regularizer. For snooker, experimentally-derived sliding friction between balls and table was reported in the range **0.178–0.245**, with **0.21** used as an average in simulations. ¹⁷

This produces:

$$\dot{\mathbf{v}}_{\parallel} = -\mu_s g \frac{\mathbf{u}}{\|\mathbf{u}\| + \varepsilon_u}, \quad \dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{r} \times \mathbf{F}_{\text{slide}})$$

with $\mathbf{r} = -R\mathbf{n}$. The coupling is the mechanism through which a sliding ball “spins up” into rolling (topspin builds from friction). ¹⁶

Closed-form sliding→rolling transition (useful for validation and as a fast integrator)

For 1D motion along direction $\hat{\mathbf{t}}$ (or for each tangential component independently), define scalar slip:

$$u = v - R\omega_t$$

For a solid sphere $I = \frac{2}{5}mR^2$ on a plane under kinetic friction μ_s , the slip decreases linearly:

$$\dot{u} = -\frac{7}{2}\mu_s g \text{sign}(u)$$

so time-to-roll is:

$$t_{\text{roll}} = \frac{2}{7} \frac{|u_0|}{\mu_s g}$$

At t_{roll} , enforce $u = 0$ (project to rolling) and continue in rolling resistance mode. This classical result is consistent with rigid-body + Coulomb friction derivations used in billiards impact modeling. ¹⁸

Rolling resistance (dominant late in motion; sets stop distance/time)

Billiards cloth exhibits measurable deceleration in the rolling regime that is well approximated as speed-independent over relevant low speeds; reported rolling deceleration values around **0.124–0.126 m/s²** correspond to an effective rolling resistance coefficient $\mu_r \approx 0.0127\text{--}0.0129$ via $a = \mu_r g$. ¹⁹

A simple and implementable model:

$$\mathbf{F}_{\text{roll}} = -\mu_r N \frac{\mathbf{v}_{\parallel}}{\|\mathbf{v}_{\parallel}\| + \varepsilon_v}$$

This gives constant deceleration $a = \mu_r g$ and thus:

$$t_{\text{stop}} = \frac{v_0}{\mu_r g}, \quad d_{\text{stop}} = \frac{v_0^2}{2\mu_r g}$$

These formulas are directly useful to match the “table speed” tests specified by WPA/BCA (ball shot with a firm stroke travels 4–4½ table lengths). ⁵

Illustrative speed–time profile (piecewise linear) using $\mu_s = 0.21$ and $\mu_r = 0.013$: the first ~0.1–0.4 s is dominated by sliding deceleration $\mu_s g$, followed by a long rolling tail with deceleration $\mu_r g$. The breakpoint timing is set by t_{roll} . ²⁰

Sidespin decay, precession, and “gyroscopic” effects

Because a sphere’s inertia is isotropic, there is no asymmetric rigid-body gyroscopic coupling (unlike a spinning top), but **precession of the spin axis** still occurs whenever the cloth/contact torque is not parallel to $\boldsymbol{\omega}$:

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1} \boldsymbol{\tau}$$

A torque component perpendicular to $\boldsymbol{\omega}$ rotates the spin axis; the instantaneous precession rate magnitude is approximately:

$$\Omega_p \approx \frac{\|\boldsymbol{\tau}_\perp\|}{I\|\boldsymbol{\omega}\|}$$

Empirically, table resistance to **sidespin** (spin about vertical) has been measured as an angular deceleration of about **22 rad/s²** in high-speed tracking experiments, giving a practical decay law:

$$\dot{\omega}_z = -\alpha_z \text{sign}(\omega_z), \quad \alpha_z \approx 22 \text{ rad/s}^2$$

for moderate $|\omega_z|$, with optional smoothing near zero to avoid jitter. ²¹

Air drag (usually secondary, but not always negligible at break speeds)

Quadratic drag:

$$\mathbf{F}_{\text{air}} = -\frac{1}{2} \rho C_D A \|\mathbf{v}\| \mathbf{v}$$

with $A = \pi R^2$. For a sphere, NASA provides background on drag behavior and coefficient variation with Reynolds number. ²²

At typical rolling speeds (~0.5–2 m/s), cloth losses dominate. At extreme break speeds (order 10 m/s), air drag can become comparable to rolling resistance in magnitude because it scales as v^2 while rolling loss is approximately constant in the simple model. ²³

Ball–ball collision physics: elastic theory, oblique impacts, rotational coupling, and impulse algorithms

Geometry and kinematics at contact

For two balls A, B of radii R_A, R_B , define:

- Line-of-centers unit normal $\mathbf{n} = \frac{\mathbf{x}_B - \mathbf{x}_A}{\|\mathbf{x}_B - \mathbf{x}_A\|}$
- Contact points relative to centers: $\mathbf{r}_A = +R_A \mathbf{n}$, $\mathbf{r}_B = -R_B \mathbf{n}$

Relative velocity at the contact point:

$$\mathbf{v}_{rel} = (\mathbf{v}_A + \boldsymbol{\omega}_A \times \mathbf{r}_A) - (\mathbf{v}_B + \boldsymbol{\omega}_B \times \mathbf{r}_B)$$

Split into normal and tangential components:

$$v_n = \mathbf{v}_{rel} \cdot \mathbf{n}, \quad \mathbf{v}_t = \mathbf{v}_{rel} - v_n \mathbf{n}$$

This decomposition underlies both impulse and compliant contact formulations. 24

Impulse-based collision response with friction (recommended for real-time engines)

Normal impulse (Newton restitution law):

$$J_n = -(1 + e) \frac{v_n}{\frac{1}{m_A} + \frac{1}{m_B}} \quad \text{for } v_n < 0$$

For spheres, rotation does not directly enter the normal effective mass because $\mathbf{r} \parallel \mathbf{n} \Rightarrow (\mathbf{r} \times \mathbf{n}) = \mathbf{0}$. 25

Tangential impulse: choose a tangential unit direction

$$\mathbf{t} = \frac{\mathbf{v}_t}{\|\mathbf{v}_t\| + \varepsilon_t}$$

Define tangential effective mass (including rotation):

$$\frac{1}{m_{eff,t}} = \frac{1}{m_A} + \frac{1}{m_B} + \mathbf{t} \cdot [\mathbf{I}_A^{-1}(\mathbf{r}_A \times \mathbf{t}) \times \mathbf{r}_A + \mathbf{I}_B^{-1}(\mathbf{r}_B \times \mathbf{t}) \times \mathbf{r}_B]$$

For equal solid spheres $I = \frac{2}{5}mR^2$, this reduces to $m_{eff,t} = m/7$, a useful check for implementation. 18

The “sticking” impulse to cancel tangential slip would be:

$$J_t^* = -\frac{\mathbf{v}_t \cdot \mathbf{t}}{1/m_{eff,t}}$$

Clamp by Coulomb friction:

$$J_t = \text{clamp}(J_t^*, -\mu_{bb}|J_n|, +\mu_{bb}|J_n|)$$

Total impulse: $\mathbf{J} = J_n \mathbf{n} + J_t \mathbf{t}$. Update:

$$\mathbf{v}_A^+ = \mathbf{v}_A^- + \frac{\mathbf{J}}{m_A}, \quad \boldsymbol{\omega}_A^+ = \boldsymbol{\omega}_A^- + \mathbf{I}_A^{-1}(\mathbf{r}_A \times \mathbf{J})$$

$$\mathbf{v}_B^+ = \mathbf{v}_B^- - \frac{\mathbf{J}}{m_B}, \quad \boldsymbol{\omega}_B^+ = \boldsymbol{\omega}_B^- - \mathbf{I}_B^{-1}(\mathbf{r}_B \times \mathbf{J})$$

This is a standard rigid-body impulse approach used in physically-based animation/graphics and is well suited to many-collision billiards. ²⁶

Realistic parameters for ball-ball impacts: restitution, friction, and energy loss

In snooker-ball collision modeling on a rough plane, ball-table sliding friction μ_s was reported as 0.178–0.245 (avg 0.21). Ball-ball friction μ_{bb} and restitution e were calibrated by matching experiments; $\mu_{bb} \approx 0.05$ and $e \approx 0.89$ minimized RMS error in their dataset. ¹⁷

That same source summarizes prior literature indicating μ_{bb} can rise substantially when balls are contaminated (e.g., chalk), with values as high as ~0.2 mentioned in the pocket-billiards literature they cite. ²⁷

Energy loss: in 1D normal impacts, the fraction of *normal relative kinetic energy* lost is:

$$\Delta E/E = 1 - e^2$$

So $e = 0.89 \Rightarrow 1 - e^2 \approx 0.21$ ($\approx 21\%$ loss in normal relative motion), while $e = 0.95 \Rightarrow \approx 9.8\%$. This is only the normal-mode loss; tangential friction can dissipate additional energy by doing work during slip. ²⁸

Contact mechanics option: Hertzian normal contact + tangential compliance

If you need physically grounded contact patch size, peak force, and contact duration, Hertzian contact provides:

$$F_n = \frac{4}{3} E^* \sqrt{R_{eff}} \delta^{3/2}$$

$$\frac{1}{E^*} = \frac{1 - \nu_A^2}{E_A} + \frac{1 - \nu_B^2}{E_B}, \quad R_{eff} = \left(\frac{1}{R_A} + \frac{1}{R_B} \right)^{-1}$$

These relations (and associated contact radius/pressure formulas) are standard in contact mechanics. ²⁹

Contact duration for Hertzian elastic sphere-sphere contact is widely used in DEM stability analysis; a common form is:

$$t_H \approx 2.94 \left(\frac{m_{eff}^2}{R_{eff} E^{*2} v_n} \right)^{1/5}$$

with $m_{eff} = (1/m_A + 1/m_B)^{-1}$. ³⁰

Using representative billiard-ball material values (phenolic resin modulus on the order of gigapascals) yields contact times on the order of **0.2–0.5 ms** for typical impact speeds (a calculation dependent on E^* , v_n , and ball sizes). ³¹

Tangential compliance (optional): Cattaneo–Mindlin-type tangential stiffness is often approximated as $k_t \propto G^* a$ (with shear modulus G^* and contact radius a), combined with a Coulomb limit $|F_t| \leq \mu F_n$. This is the theoretical basis behind many DEM tangential “spring + Coulomb cap” laws. ³²

Speed-dependent restitution (optional refinement)

Viscoelastic impact theory predicts that restitution decreases with impact speed, often with characteristic $v^{1/5}$ scaling for Hertzian viscoelastic spheres; coefficients can be fit to experiments or taken from viscoelastic-sphere literature. ³³

For billiards simulation, a pragmatic engineering approach is to implement $e(v_n)$ as a fitted curve (e.g., piecewise linear in $\log v_n$ or a viscoelastic-Hertz form) and validate against measured rebound speeds across a range of cue-ball speeds. ³⁴

Ball-cushion and table-edge collisions: geometry, restitution, friction, spin transfer, and special tables

Cushion geometry constraints (pool)

The cushion’s geometric placement affects (i) rebound angle, (ii) hop propensity, and (iii) “short/long” bank behavior. WPA recommends cushion nose height (nose-line to bed) of **63½% ± 1% of ball diameter**, i.e., **62½%–64½%**. For a 57.15 mm pool ball, this implies a nose height near 36.3 mm. ³⁵

WPA also specifies a cloth-covered cushion width (featherstrip edge to nose) between **1⅞” and 2”** and includes a performance criterion: a ball shot from the head spot through the foot spot with a firm stroke travels **4 to 4½ table lengths without jumping**. ³⁶

Cushion impact physics and parameter values from experiments

Ball-cushion collisions are strongly affected by friction and spin (topspin/sidespin). In a theoretical + experimental calibration study of cushion impacts (snooker context), the best-fit parameters under a “negligible cushion deformation” assumption were:

- coefficient of restitution (their fitted value): **0.98**
- sliding coefficient of friction ball-cushion: **0.14** ³⁷

They also note a practical limit: deviation for incident speeds > **2.5 m/s** suggests the rigid-cushion assumption becomes less valid above that speed. ³⁸

The same source reports an experimentally obtained representative restitution value around **0.818** for a rolling ball perpendicularly colliding with the cushion, emphasizing that it incorporates friction and 3D impulse effects rather than being a “pure” normal restitution. ³⁹

Implementable impulse model for ball-cushion (sphere vs. surface)

Model the cushion face (locally) as a plane or ruled surface (depending on profile). At contact:

- compute surface normal $\mathbf{n}$ at contact point,
- compute relative contact-point velocity $\mathbf{v}_{rel} = \mathbf{v} + \boldsymbol{\omega} \times \mathbf{r} - \mathbf{v}_{surf}$ (usually $\mathbf{v}_{surf} = 0$),
- solve for normal impulse with restitution using the full effective mass:

$$J_n = -(1 + e_c) \frac{v_n}{\frac{1}{m} + \mathbf{n} \cdot [\mathbf{I}^{-1}(\mathbf{r} \times \mathbf{n}) \times \mathbf{r}]}$$

Unlike ball-ball, here $\mathbf{r} \times \mathbf{n}$ is generally nonzero because the contact point is not exactly on the line joining centers (the cushion normal is not radial), so rotation affects the normal impulse. ⁴⁰

Tangential impulse is handled as in ball-ball using a Coulomb cap with μ_{bw} (ball-cushion friction), and the updates follow the same $\Delta \mathbf{v}$, $\Delta \boldsymbol{\omega}$ equations. ⁴¹

Table shapes and variants

Rectangular pool tables: 6 pockets, corner/side pocket geometries differ (see pocket section). Geometry is spec-driven by mouth width, shelf depth, and cut angles. ⁴

Carom tables: no pockets, but table size and cushion height are specified. UMB rules specify a playing surface of **2.84 m × 1.42 m** (± 5 mm) and cushion height at the frontmost point of **37 mm ± 1 mm**, with tournament tables using heating to reduce humidity effects. ¹⁴

Snooker tables: WPBSA specifies playing area (3569×1778 mm) and ball diameter 52.5 mm, but pocket openings are required to conform to authorized templates (so software should treat them as a “template geometry,” not a single mouth-width number). ¹²

Oval/elliptical and novelty tables: treat as changes in boundary geometry only; collision physics is unchanged but the cushion normal field $\mathbf{n}(\mathbf{x})$ varies continuously, so robust surface contact and reflection are more important. (No single governing spec; treat as user-provided geometry.)

Ball-pocket physics: pocket geometry, entry, rattle/rebound, and escape

Pocket geometry parameters from WPA and BCA specifications

For pool tables, WPA recommended specs define:

- **Corner pocket mouth:** 4.5”–4.625”
- **Side pocket mouth:** 5.0”–5.125”
- **Vertical pocket angle (back draft):** 12°–15°
- **Cut angles:** corner entrance 142° (+1), side entrance 104° (+1)
- **Shelf depth:** corner 1”–2¼”; side 0”–0.375”

- **Facings thickness:** min $1/16''$, max $1/4''$ (preferred max $1/8''$), with facings somewhat harder than cushions 35

BCA (recreational/non-professional) specs are looser in mouth widths:

- **Corner pocket mouth:** $4\frac{7}{8}''$ – $5\frac{1}{8}''$
- **Side pocket mouth:** $5\frac{3}{8}''$ – $5\frac{5}{8}''$
- **Vertical pocket angle:** 12° – 15°
- **Corner shelf:** $1\frac{5}{8}''$ – $1\frac{7}{8}''$; **Side shelf:** $0''$ – $\frac{3}{8}''$ 42

These dimensions heavily determine the “acceptance envelope” for cut shots—especially shelf depth and jaw angles which govern whether the ball’s center of mass can pass the pocket fall line before striking a facing. 4

A physically grounded way to model pocketing vs. rebound

A pocket is best treated as a **3D funnel + gravity capture** problem:

1. Geometry representation

Represent the pocket region as a set of collision surfaces:

2. cushion/facing “jaws” (rubber),
3. slate cut/bevel surfaces (hard),
4. liner/boot surfaces (rubber/leather/plastic per WPA),
plus an “open volume” beyond the slate cut where the ball can fall. 36

5. Entry condition (mouth crossing)

The ball is “in the pocket mouth” when its sphere intersects the mouth plane and its center lies within the 2D mouth aperture expanded by ball radius (equivalently, use signed distance fields). WPA defines mouth measurement between cushion noses at the direction change into the pocket. 4

6. Point-of-no-return criterion (shelf + gravity)

Define a “shelf plane” located at the slate vertical cut (or at an equivalent geometrically defined drop boundary). WPA defines shelf as distance from the mouth line to the vertical cut of the slate pocket cut. 36

A robust pocketing rule is:

- If the ball center crosses beyond the shelf boundary and the local gravitational potential gradient points “down into” the pocket cavity, then treat as **captured** unless a collision produces upward/outward velocity sufficient to reverse beyond the shelf.

Practically: once $\mathbf{x}$ enters a “capture volume” (a region where no continuous path exists back to the table without intersecting geometry), you can switch the ball to pocket dynamics only (gravity + pocket-wall contacts) and stop simulating cloth contact.

1. Rebound and rattle (jaw impacts)

If the ball collides with jaw/facing surfaces before passing the shelf, treat it as an ordinary collision

with the facing surface using appropriate restitution/friction. WPA specifies facing materials and thickness constraints (important because thick/soft facings enlarge dissipation and change rattle behavior). ³⁵

2. Escape after partial entry

Escape occurs when the ball's center remains on the table side of the shelf boundary and its post-collision velocity has outward component large enough to clear the facing geometry. This is inherently geometry- and speed-dependent; in simulation, it “falls out” naturally if the pocket is modeled as real surfaces and you avoid premature capture triggers.

Impact forces and trajectories inside the pocket

Once a ball is inside the pocket cavity:

$$m\dot{\mathbf{v}} = m\mathbf{g} + \sum_k \mathbf{F}_{contact,k}$$

Because pocket liners are specified to direct a ball downward when struck below the rail rim, they should be modeled as sloped/curved surfaces with significant energy loss (lower restitution, higher damping) compared to cushion rebound surfaces. ³⁶

For practical simulation, use:

- lower restitution (e.g., 0.2–0.5) and higher friction for liner/boot contacts than for cushion nose contacts, and validate by observing whether “trampoline returns” are rare, consistent with WPA's requirement that ball returns not bounce balls back onto the table. ³⁶

Parameters and defaults: realistic values, recommended table, timestep guidance, and validation experiments

Regulation-driven geometry and ball properties

Pool balls (WPA): cast phenolic resin, diameter 2¼" (+0.005") and mass 5½–6 oz (156–170 g). ⁴³

Snooker balls (WPBSA): diameter 52.5 mm ± 0.05 mm; ball set weight tolerance within 3 g (snooker). ¹²

Carom balls (UMB): diameter 61–61.5 mm (document text shows “cm,” but context is ball diameter) and mass 205–220 g; cushion height at frontmost point 37 mm ± 1 mm. ¹⁴

Ball inertia (homogeneous sphere default):

$$I = \frac{2}{5}mR^2$$

This is a good engineering approximation for billiard balls unless you have a measured inertia tensor. ⁴⁴

Material properties needed for contact mechanics models

If you implement Hertzian/compliant contact, you need E , ν for balls and cushions:

- Phenolic laminate datasheets report Young's modulus on the order of **6.3 GPa** (example paper/phenolic laminate). ⁴⁵
- Phenolic molding compound datasheets report Poisson's ratio in tension **~0.34–0.36**. ⁴⁶
- Rubber-like materials are often treated as nearly incompressible, with Poisson's ratio frequently measured around **0.46–0.49** in elastomers (example: polychloroprene rubbers). ⁴⁷
- Tournament cushion manufacturers specify natural-rubber cushion hardness around **40–45 Shore A** for certain billiard cushion products, supporting the need to treat cushions as compliant and dissipative relative to balls. ⁴⁸

These values are sufficient to compute effective modulus E^* and contact time scales for stability analysis and damping calibration. ⁴⁹

Coefficients of friction and restitution: experimentally grounded defaults

- Ball-table sliding friction: $\mu_s \in [0.178, 0.245]$, recommend default **0.21** (snooker; often similar order for pool cloth; calibrate per cloth). ¹⁷
- Ball-ball friction: $\mu_{bb} \approx 0.05$ (clean), can be higher with contamination; treat as tunable. ¹⁷
- Ball-ball energetic restitution: calibrated around **0.89** in the cited snooker collision study; other billiards sources report higher values, so treat e as speed- and cleanliness-dependent and validate experimentally. ²⁸
- Ball-cushion: fitted $e \approx 0.98$, $\mu_{bw} \approx 0.14$ under a stiff-cushion assumption, with validity limits around 2.5 m/s incident speed. ³⁷
- Sidespin decay: angular deceleration $\alpha_z \approx 22 \text{ rad/s}^2$. ⁸

Typical speeds and impulses for validation targets

- Break shot cue ball speeds from video analysis were reported in the range **22–26 mph** (avg **24 mph**), i.e., roughly 9.8–11.6 m/s (avg ~10.7 m/s). ⁵⁰
- High-end cue striking speed in a snooker robotic/human comparison was reported up to about **2.75 m/s** as a “typical high-end cue velocity” in play. ⁵¹

A first-order cue impulse estimate (ignoring cue compliance) is:

$$J \approx m \Delta v$$

So for $m \approx 0.17 \text{ kg}$ and $\Delta v \approx 10 \text{ m/s}$, $J \approx 1.7 \text{ N}\cdot\text{s}$. For a moderate 2 m/s shot, $J \approx 0.34 \text{ N}\cdot\text{s}$. These are useful sanity checks when relating cue-tip force profiles to ball launch speeds.

⁵²

Numerical stability, contact detection tolerances, and timestep guidance

Collision detection (CCD is strongly recommended):

- Ball-ball time of impact can be solved analytically for spheres via quadratic root of $\|\mathbf{x}_{rel} + t\mathbf{v}_{rel}\| = R_A + R_B$.
- Ball-plane and ball-cushion can be solved similarly as ray/sphere vs plane; for curved cushions and pockets use conservative advancement or signed-distance fields.
Impulse-based simulation work emphasizes robust detection to prevent interpenetration and instability. ⁵³

Penetration slop / tolerances:

Use a small spatial slop ε_p (e.g., 10–50 μm in world units if using meters; larger if using centimeters) for contact persistence and to avoid jitter, plus velocity thresholds for “sleeping” balls near rest. This is standard practice in iterative contact solvers. ⁵⁴

Time stepping strategies:

- **Event-driven (recommended for billiards):** integrate continuous dynamics with a fixed (or adaptive) Δt but resolve all collisions by advancing to earliest time-of-impact and applying impulses, potentially multiple times per frame. This avoids tunneling even at high break speeds. ⁵⁵
- **Time stepping with constraints:** if using a projected Gauss–Seidel (PGS) style solver, use multiple iterations per frame for stability, and clamp friction impulses to a Coulomb cone. ⁵⁴
- **Compliant/Hertz DEM-style:** stability requires Δt small relative to the Hertz contact duration t_H ; DEM literature often relates stable/critical time steps to fractions of t_H . ³⁰

Summary table of recommended default parameters

The table below provides a practical default set (pool-centric) that is consistent with published equipment specs and experimental ranges. Values should be tuned to a specific table/cloth/ball set via the validation experiments listed afterward.

Category	Parameter	Recommended default	Notes / source basis
Ball geometry	Diameter D	57.15 mm	WPA pool ball spec ⁵⁶
	Radius R	28.575 mm	Derived from D ⁵⁶
Ball mass	m	0.170 kg	Upper end of WPA 156–170 g range (choose mid/upper for stability) ⁴³
Ball inertia	I	$0.4mR^2$	Homogeneous sphere approximation ⁵⁷
Ball material (optional Hertz)	E	6.3 GPa	Representative phenolic laminate modulus ⁴⁵
	ν	0.35	Typical phenolic Poisson ratio range center ⁴⁶

Category	Parameter	Recommended default	Notes / source basis
Cloth friction	Sliding μ_s	0.21	Avg reported in snooker modeling; calibrate for pool cloth ¹⁷
	Rolling μ_r	0.013	From reported rolling deceleration (effective) ¹⁹
Spin decay	α_z	22 rad/s ²	Sidespin resistance measurement ⁸
Ball-ball impact	Restitution e_{bb}	0.90	Start near 0.89 fit; allow speed-dependence ²⁸
	Friction μ_{bb}	0.05	Clean balls; increase for “chalky” conditions ¹⁷
Ball-cushion impact	Restitution e_{bw}	0.95–0.98	0.98 fitted in cushion-impact study (stiff-cushion regime) ³⁸
	Friction μ_{bw}	0.14	Fitted value ⁵⁸
Cushion geometry	Nose height	0.635·D	WPA: 63½% ± 1% of ball diameter ⁵⁹
Pockets (pool)	Corner mouth	4.50–4.625 in	WPA recommended ³⁶
	Side mouth	5.00–5.125 in	WPA recommended ³⁶
	Corner shelf	1.0–2.25 in	WPA recommended ³⁶
	Side shelf	0–0.375 in	WPA recommended ³⁶

Suggested validation experiments (to lock parameters to a specific table)

Each experiment is chosen to isolate a small subset of parameters so you can fit them cleanly.

1. Rolling resistance fit (cloth/table speed)

Launch a ball in near-pure rolling at several initial speeds (0.5–2 m/s), record $v(t)$, fit constant deceleration a and compute $\mu_r = a/g$. Compare with reported rolling deceleration scale (~0.124–0.126 m/s²) and WPA/BCA “4–4½ table lengths” behavior. ⁶⁰

2. Sliding friction fit (skid phase)

Launch with known initial spin/velocity mismatch (e.g., stun/draw/follow regimes) and fit μ_s via the analytical $t_{\text{roll}} = \frac{2}{7} \frac{|u_0|}{\mu_s g}$ relation. Target μ_s in the 0.178–0.245 band unless your cloth differs. ¹⁷

3. Sidespin decay test

Spin the cue ball about vertical while nearly stationary, track $\omega_z(t)$, and fit α_z (constant decel) or a viscous term. Compare to ~22 rad/s². ⁸

4. Ball-ball restitution/friction calibration

Perform head-on and controlled cut-angle collisions at several speeds, measure outgoing speeds and angular velocities, and fit $e_{bb}(v_n)$ and μ_{bb} . Use $\mu_{bb} \approx 0.05$ and $e \approx 0.89$ as starting values, but validate against your balls/cleanliness. ²⁸

5. Ball-cushion coefficient fit

Shoot a rolling ball into a cushion at varying incident angles and modest speeds (≤ 2.5 m/s if using stiff-cushion model assumptions), and fit e_{bw} and μ_{bw} by minimizing rebound speed/angle error. Use 0.98 and 0.14 as starting values. ³⁷

6. Pocket acceptance mapping

Systematically vary approach angle, speed, and spin into each pocket type (corner vs side), and record pocketed vs rejected outcomes. Use WPA/BCA geometric parameters (mouth, shelf, cut angles, back draft) as the baseline geometry and fit liner/facing restitution/friction to reproduce rattle-outs and “hangups.” ⁴

1 4 5 11 35 36 43 52 56 59 Document3

<https://wpapool.com/wp-content/uploads/2024/01/RECOMMENDED-EQUIPMENT-SPECIFICATIONS.pdf>

2 6 9 16 17 20 27 28 34 https://drdavepoolinfo.com/physics_articles/Mathavan_Sports_2014.pdf

https://drdavepoolinfo.com/physics_articles/Mathavan_Sports_2014.pdf

3 7 15 18 24 25 26 40 44 53 55 57 <https://people.eecs.berkeley.edu/~jfc/mirtich/thesis/>

[mirtichThesis.pdf](https://people.eecs.berkeley.edu/~jfc/mirtich/thesis/mirtichThesis.pdf)

<https://people.eecs.berkeley.edu/~jfc/mirtich/thesis/mirtichThesis.pdf>

8 10 21 51 https://irep.ntu.ac.uk/id/eprint/31391/1/PubSub8899_Mathavan.pdf

https://irep.ntu.ac.uk/id/eprint/31391/1/PubSub8899_Mathavan.pdf

12 13 1819_WPBSA - Rulebook Design A6 Text.indd

<https://wpbsa.com/wp-content/uploads/Rulebook-Website-Updated-May-2022-2.pdf>

14 Microsoft Word - 7914A603.doc

https://files.umb-carom.org/public/rules/carom_rules.pdf

19 23 60 https://www.researchgate.net/figure/The-ball-rolling-on-the-table-It-shows-the-forces-that-are-acting-on-the-ball-while-it_fig5_228372702

https://www.researchgate.net/figure/The-ball-rolling-on-the-table-It-shows-the-forces-that-are-acting-on-the-ball-while-it_fig5_228372702

22 <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/drag-of-a-sphere/>

<https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/drag-of-a-sphere/>

29 49 https://www.researchgate.net/profile/Abdelkader-Bouaziz/post/How_to_set_load_for_load-controlled_nanoindentation_tests/attachment/5ef0ded8c2917800015f93b6/AS%3A905267466227718%401592843991905/download/

[Introduction%2Bto%2BContact%2BMechanics_Anthony%2BC.%2BFischer-Cripps.pdf](https://www.researchgate.net/profile/Abdelkader-Bouaziz/post/How_to_set_load_for_load-controlled_nanoindentation_tests/attachment/5ef0ded8c2917800015f93b6/AS%3A905267466227718%401592843991905/download/)

https://www.researchgate.net/profile/Abdelkader-Bouaziz/post/How_to_set_load_for_load-controlled_nanoindentation_tests/attachment/5ef0ded8c2917800015f93b6/AS%3A905267466227718%401592843991905/download/

[Introduction%2Bto%2BContact%2BMechanics_Anthony%2BC.%2BFischer-Cripps.pdf](https://www.researchgate.net/profile/Abdelkader-Bouaziz/post/How_to_set_load_for_load-controlled_nanoindentation_tests/attachment/5ef0ded8c2917800015f93b6/AS%3A905267466227718%401592843991905/download/)

30 31 https://www.research.ed.ac.uk/files/81027966/Burns_et_al_2019_IJNME_Postprint.pdf
https://www.research.ed.ac.uk/files/81027966/Burns_et_al_2019_IJNME_Postprint.pdf

32 <https://www.sciencedirect.com/science/article/abs/pii/S0167663698000647>
<https://www.sciencedirect.com/science/article/abs/pii/S0167663698000647>

33 The oblique impact of elastic spheres
https://www.sciencedirect.com/science/article/abs/pii/S0043164876902015?utm_source=chatgpt.com

37 38 39 41 58 https://drdavepoolinfo.com/physics_articles/Mathavan_IMechE_2010.pdf
https://drdavepoolinfo.com/physics_articles/Mathavan_IMechE_2010.pdf

42 BCA Equipment Specifications
https://cdn.ymaws.com/bca-pool.com/resource/resmgr/imported/BCAEquipmentSpecifications_2008.pdf

45 Material Data Sheet
https://www.directplastics.co.uk/pub/pdf/datasheets/Tufnol%20Data%20Sheet.pdf?utm_source=chatgpt.com

46 Plenco's 02000 series
https://plenco.com/sheets/02000-General-Purpose-Phenolic-Materials-Plenco-June2018.pdf?utm_source=chatgpt.com

47 Elastic Properties of Polychloroprene Rubbers in Tension and ...
https://pmc.ncbi.nlm.nih.gov/articles/PMC7602244/?utm_source=chatgpt.com

48 Tournament billiard cushions for a perfect play
https://artemis-kautschuk.de/en/products/billiard-cushions/?utm_source=chatgpt.com

50 https://drdavepoolinfo.com/resource_files/Onoda_break_speed_BD_article.pdf
https://drdavepoolinfo.com/resource_files/Onoda_break_speed_BD_article.pdf

54 https://box2d.org/files/ErinCatto_IterativeDynamics_GDC2005.pdf
https://box2d.org/files/ErinCatto_IterativeDynamics_GDC2005.pdf